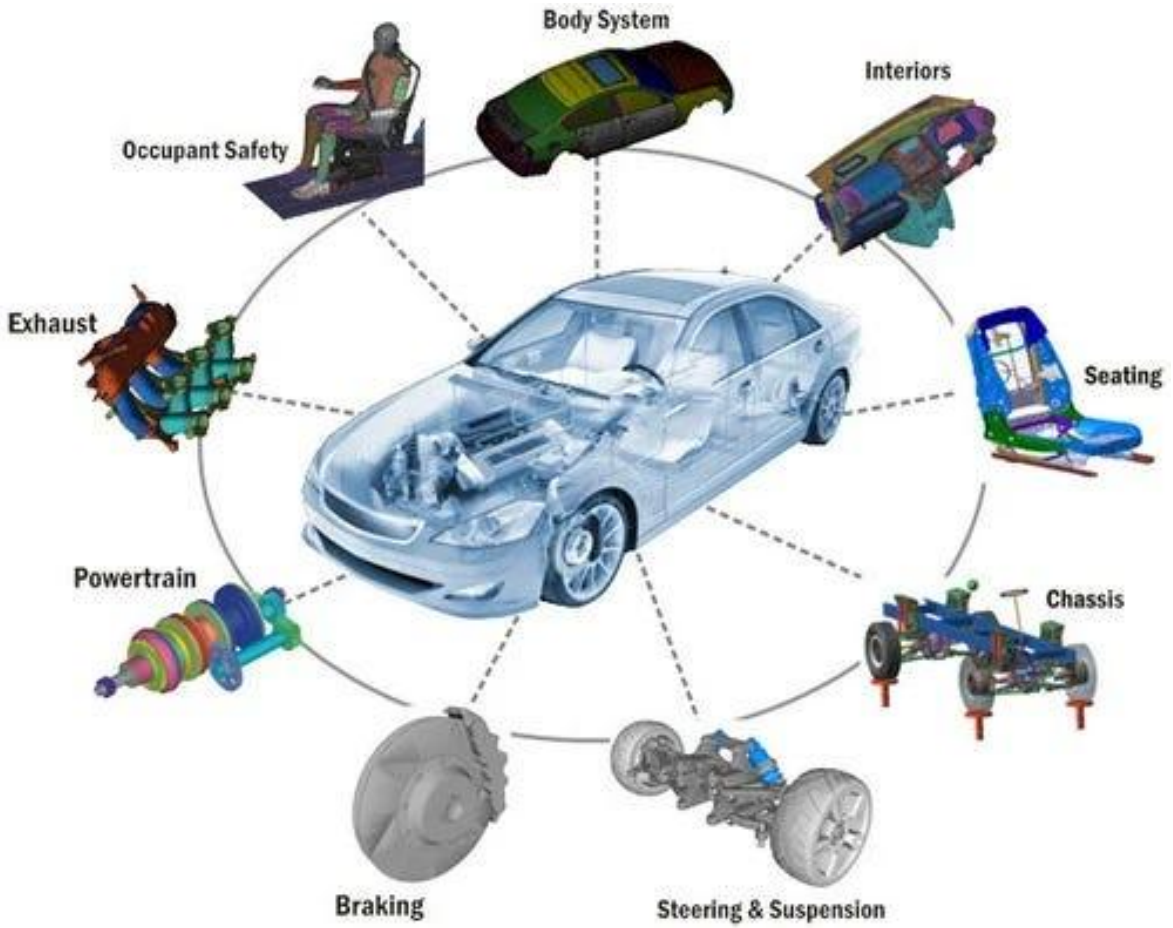
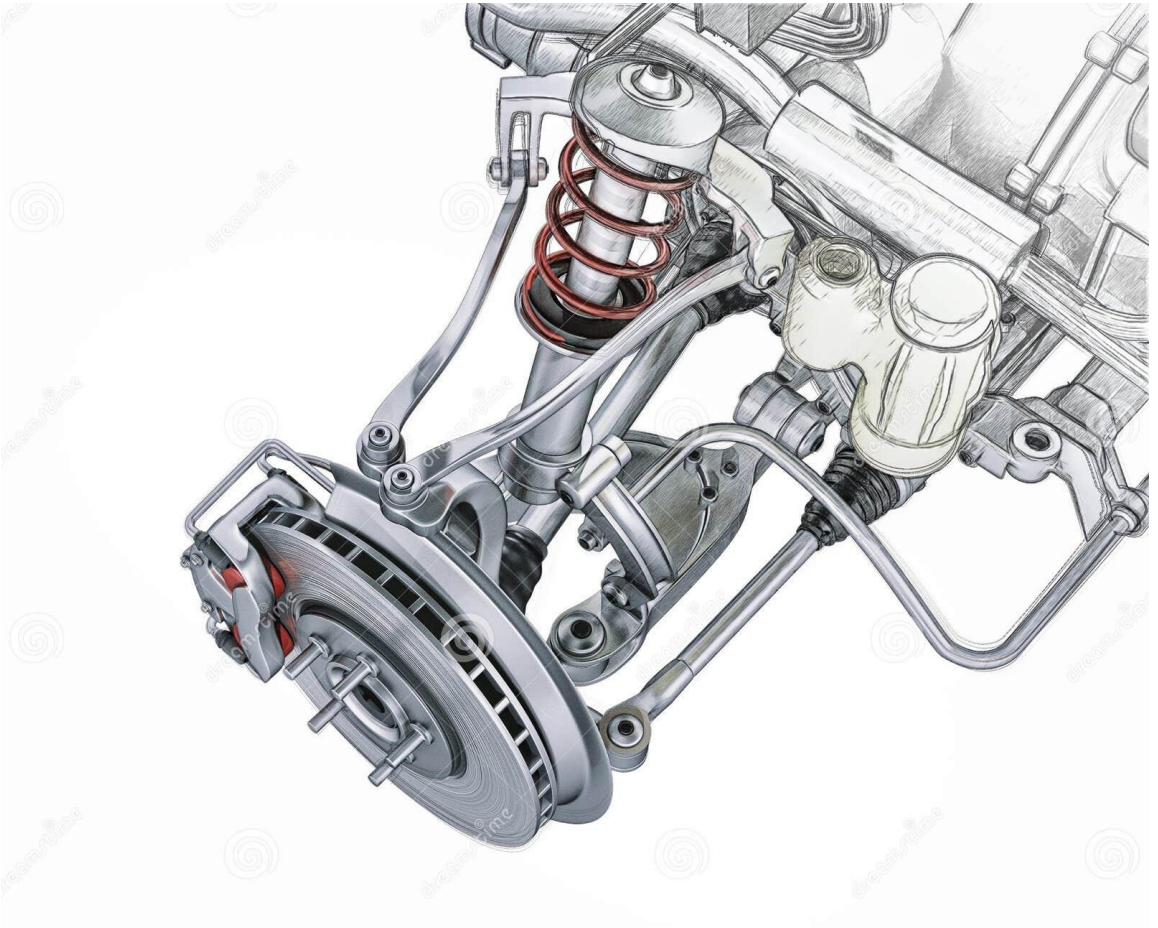




Automotive Engineering (MAM 336)

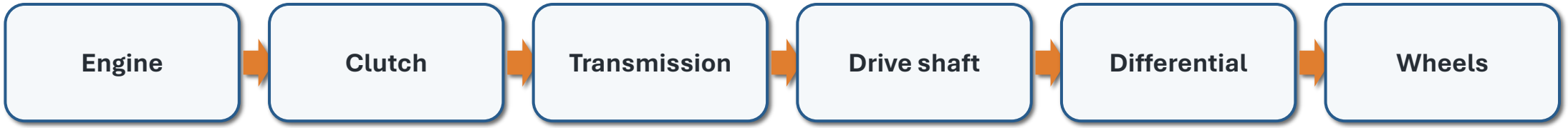
Week 5

Gear Box

DR/ MOHAMED EL TAHER MOHAMED

Power Flow Through the Vehicle

The drive train indicates the power flow from the engine to the driving wheels.



Power source

Torque / speed conversion

Torque distribution

Introduction

The road conditions are not uniform whenever we are in travel in the vehicle. Moreover we do not drive the vehicle always at the same speed. The vehicle must be operated at lower speed when it needs more pulling power. Moreover based on the operating conditions, the vehicle requires different speed and torque. To perform this, a box containing different gears, shafts and bearings is required. This box is called as the gear box. It can be also called as teeth wheel box. By using this box, the driver can achieve different speeds and torques for a vehicle. Gear box is the one which comprises of various teethed gears for different vehicle speed conditions.

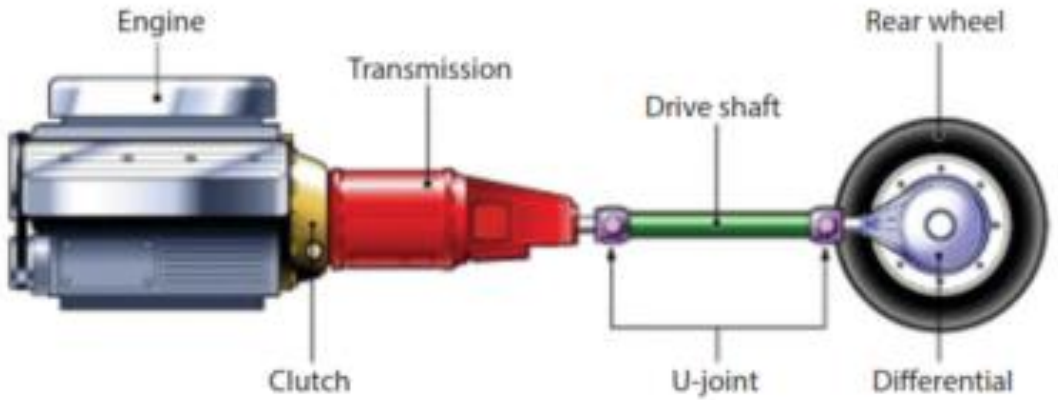


Figure 1.7 Transmission system.

Gear box is located at the second place of the vehicle's transmission system. In the vehicle after the engine, clutch is located. After the clutch gear box is located. It is located in between the clutch and the propeller shaft.

Sliding mesh gear box

Advantages of sliding mesh gearbox

1. Simple in construction
2. Simple in maintenance
3. Suitable for heavy vehicles
4. Low cost
5. Higher mechanical efficiency
6. Production cost is low

Disadvantages

1. Gear changing is difficult
2. Noisy in operation
3. Experience is required for changing gear
4. The edges of teeth wear rapidly.

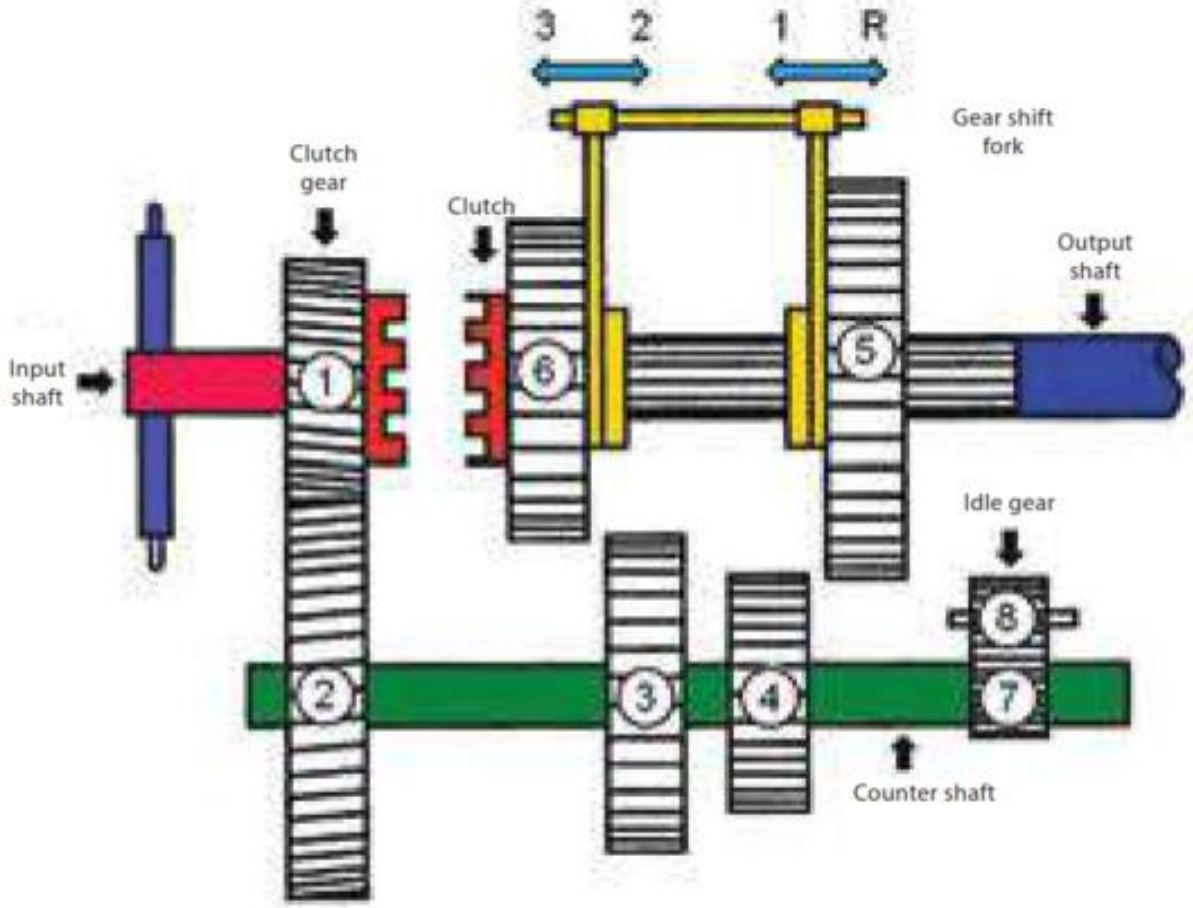


Figure 3.5.1 Sliding mesh gear box

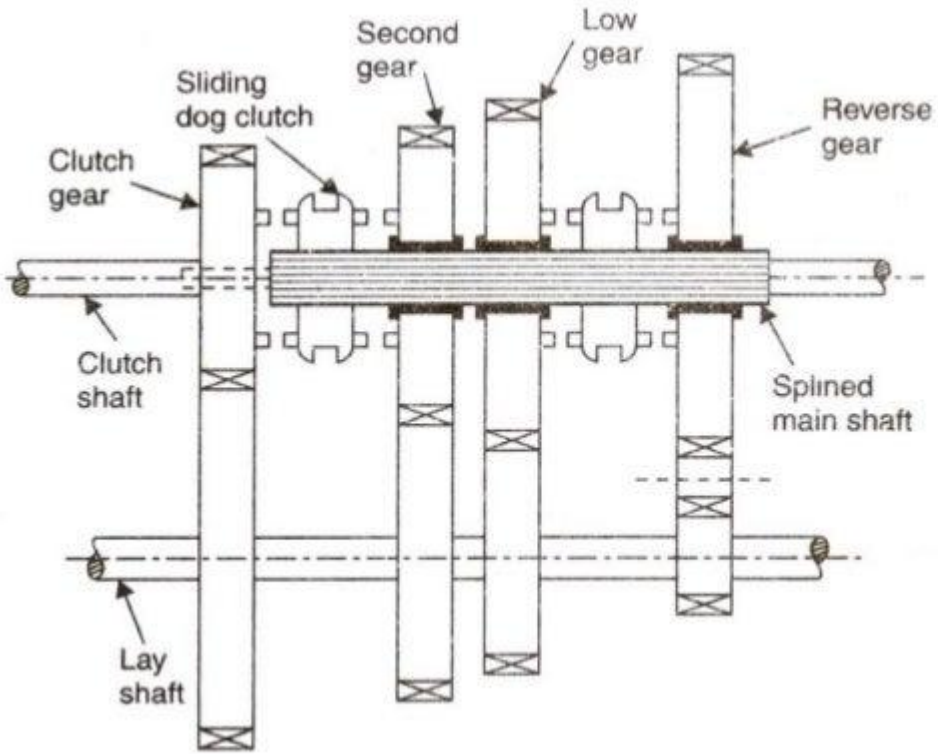
Constant mesh gear box

Advantages of constant mesh gear box

1. Gear changing is simple
2. Less noise
3. Less chance for the teeth of the gears to break up.
4. Smoothness in operation
5. Low maintenance

Disadvantages

1. For changing gears from one speed to other speed, double de clutching has to be done.
2. More wear
3. Possibility of wear in dog clutch.



Synchromesh gear box

Synchromesh gear box is similar in construction as constant mesh gear box but differs slightly. Instead of dog clutch used in constant mesh gear box, a synchronizing unit is used in synchromesh gear box. In the synchromesh mesh gear box the gears 6, 7 and 8 in the output shaft are always in mesh with the gears 5, 4 and 3 in the lay shaft. In the clutch shaft and in the lay shaft there are gears 1 and 2 respectively which are always in mesh with each other. The gears in the main shaft are supported by bushes. Gear 5 in the lay shaft is in mesh with an idler gear. After the clutch gear in the clutch shaft and before the reverse gear in the main shaft there are synchronizers (S_2 and S_1 respectively) present. These synchronizers are allowed to move on the slots made on the main output shaft. All the shafts in the gear box are supported by the bearings in the housing.

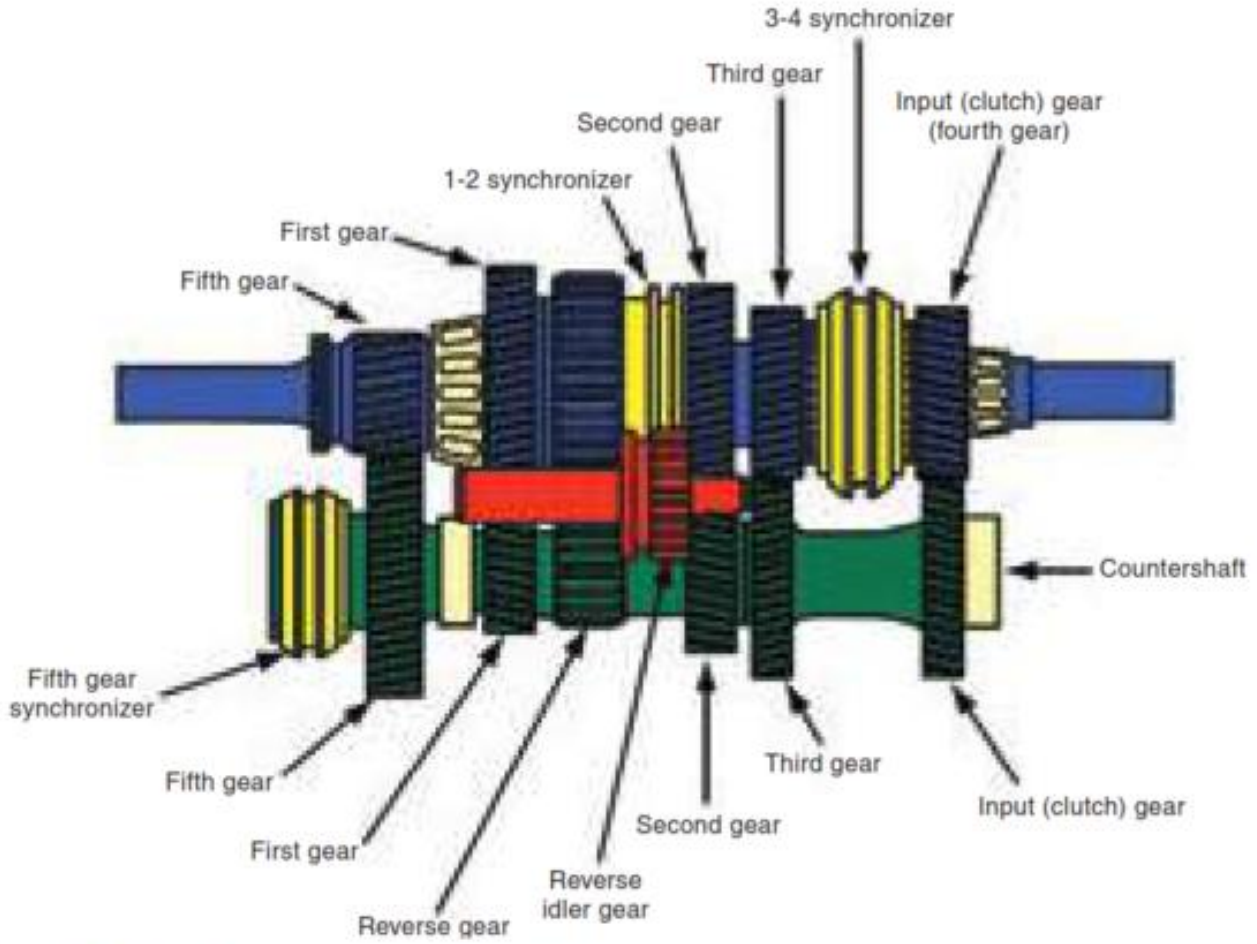


Figure 3.5.3 (a) A view of gear arrangements of a 5 speed synchromesh gear box

Synchromesh gear box

Neutral position

During neutral condition both the synchronizers (S_2 and S_1) are not in mesh with any gears as shown in Figure.

All gears in the shafts 1,2& 3,4 & 5,6 & 7,8,9 rotate. However the main output shaft does not rotate as the dog clutches are not in engagement with any of the gears. Hence rotational power is not transmitted.

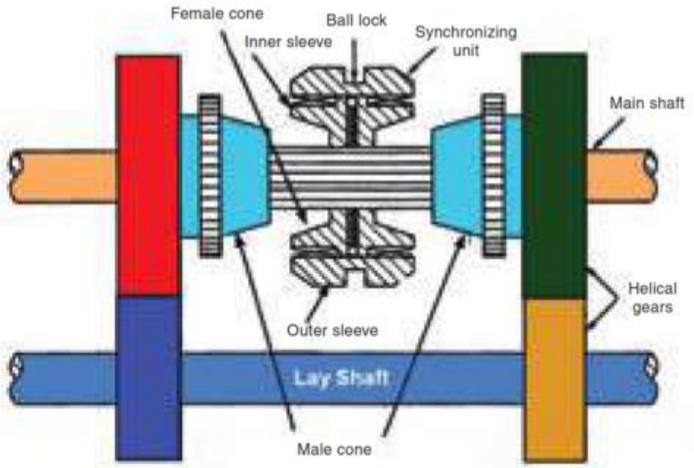


Figure 3.5.3 (c) Layout of a synchromesh gear box

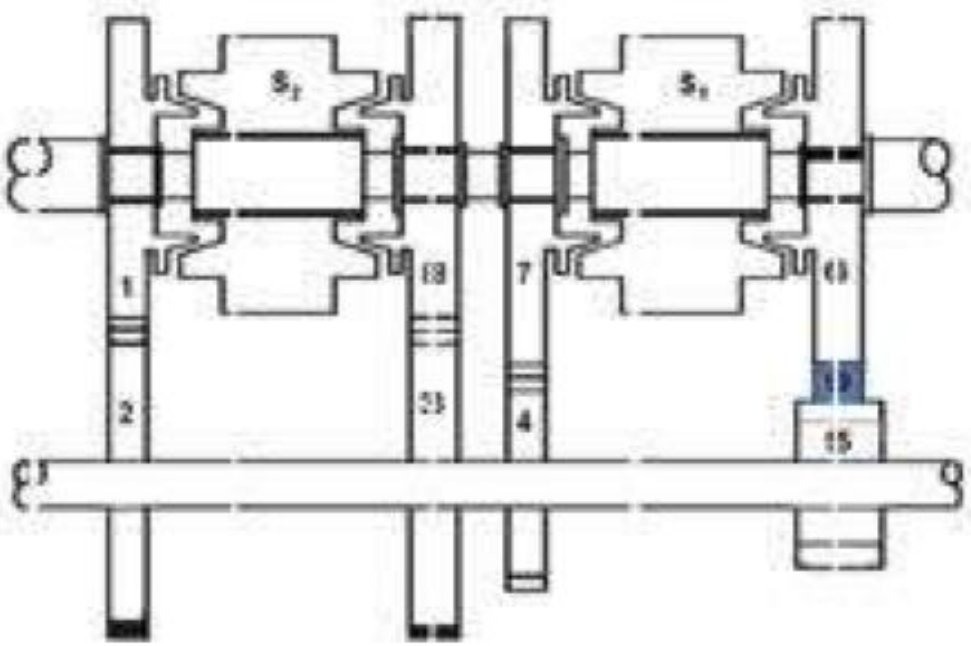


Figure 3.5.3.1 Neutral condition S1 and S2 are held stationary

Synchromesh gear box

First gear

By actuating the gear shift lever, the synchronizer S_1 is moved towards left side as shown in Figure 3.5.3.2 to engage gear 7 in the main shaft.

Now the rotational power is transmitted to the main shaft through the dog clutch. Now the speed is low. The power transmission from the clutch shaft to the output shaft is by gear 1-2 is means rotational power is transmitted to main shaft by 1-2-4-7-S_1.

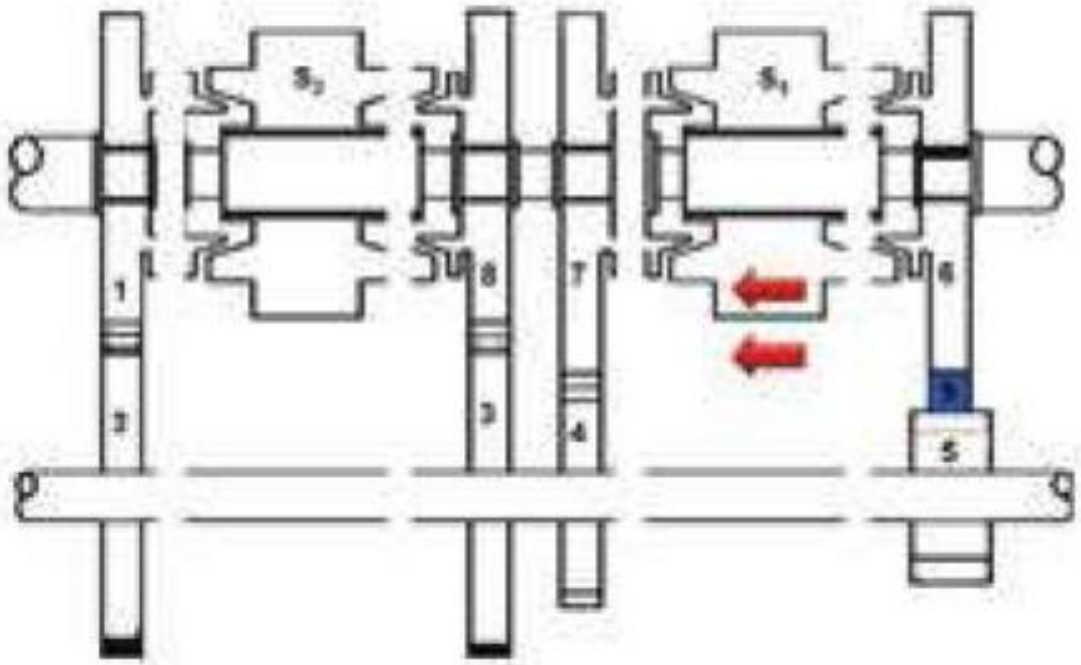


Figure 3.5.3.2 First gear condition

Synchromesh gear box

Second gear

For obtaining the second gear, by actuating the gear shift lever, the synchronizer S_2 is moved towards right to engage the gear 8 in the output main shaft. Now the rotational power is transmitted to the main shaft through the synchronizer S_2. Now the speed of the main shaft is higher than the speed of the first gear. The power transmission from the clutch shaft to the output shaft is through the gears 1-2-3-8-S_2.

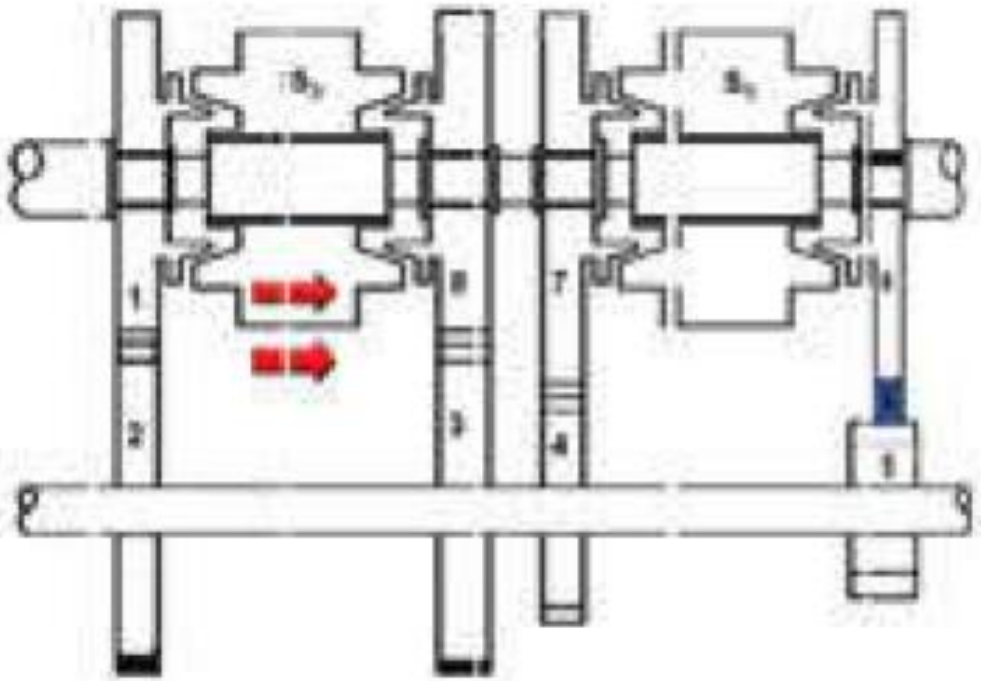


Figure 3.5.3.3 Second gear condition

Synchromesh gear box

Third gear

To achieve the third gear the synchronizer S_2 is moved left to engage with the clutch shaft gear directly. Now the output shaft rotates at the speed same as the clutch shaft speed. The rotational power is transmitted to the main shaft directly through the gear 1-S_2.

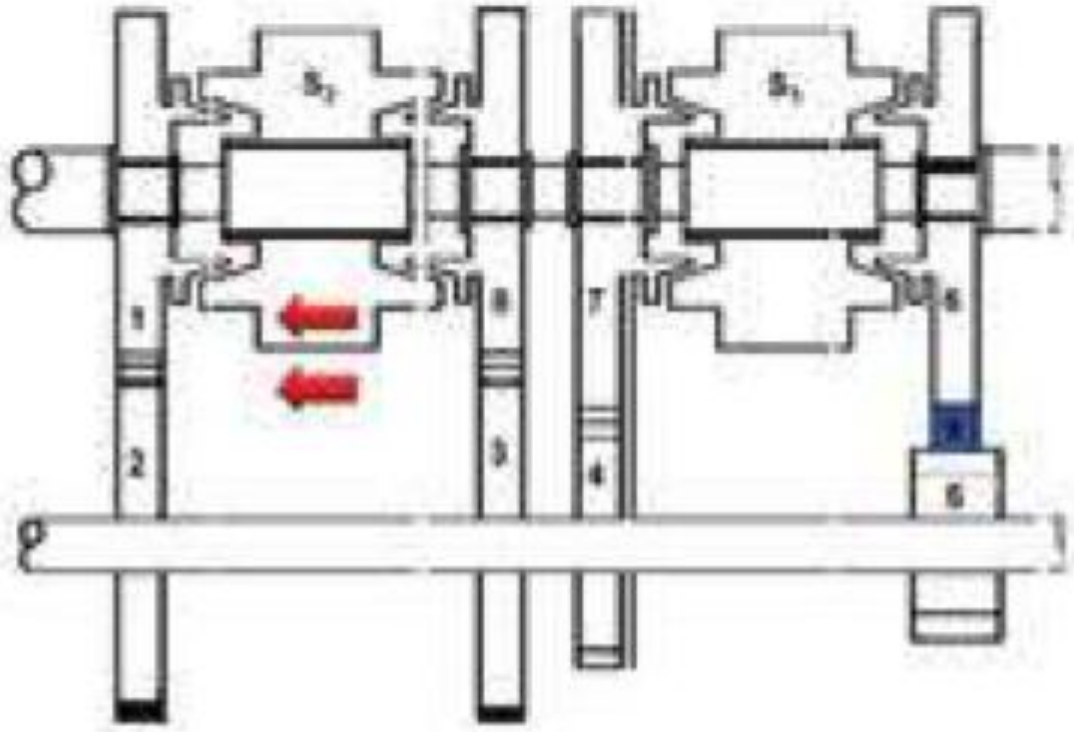


Figure 3.5.3.4 Third gear condition

Synchromesh gear box

Reverse Speed

To obtain the reverse gear, the synchronizer S_1 is actuated to move right towards the reverse gear 6 to engage as shown in Figure 3.5.3.5. As the idler gear is placed between the gears 5 in the lay shaft and 6 in the main shaft the idler gear changes the direction of rotation of the gear 6 in the main shaft. Now the vehicle moves in the reverse direction. The power transmitted from the clutch shaft to the main shaft is through gears 1-2-5-Idler-6-S_1.

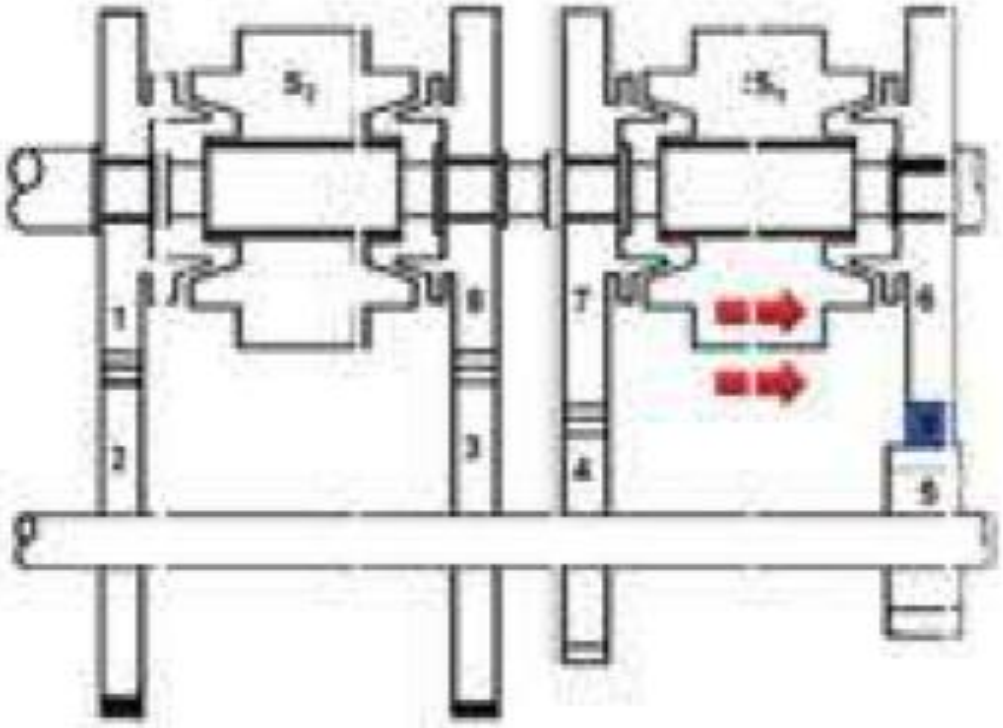
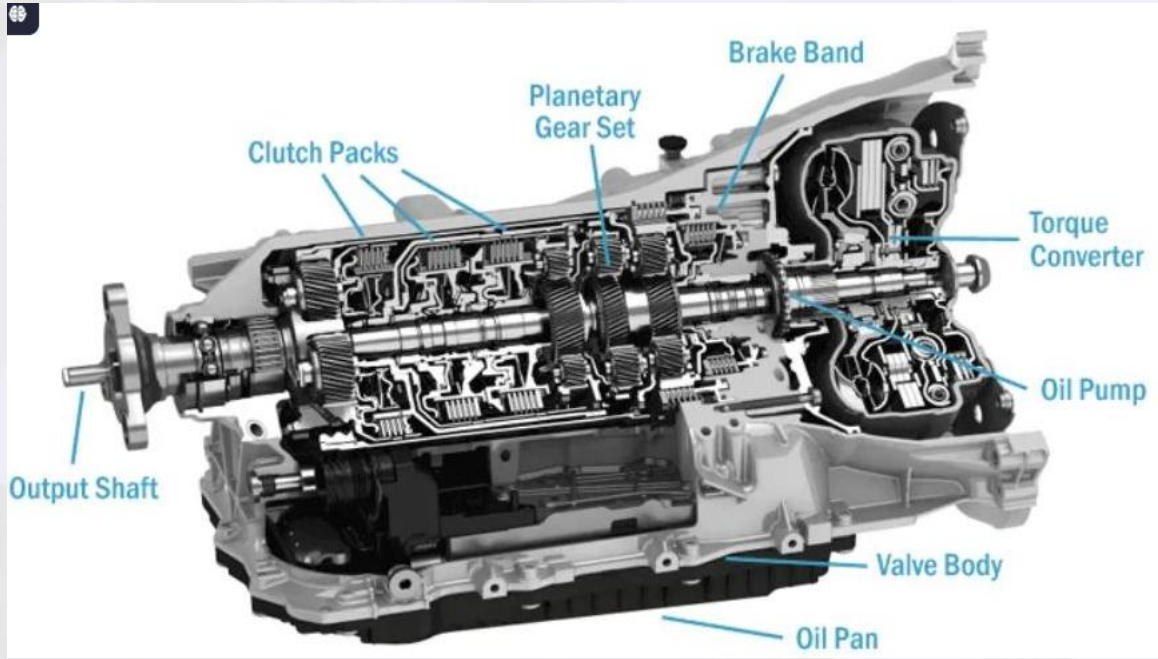


Figure 3.5.3.5 Reversed gear condition

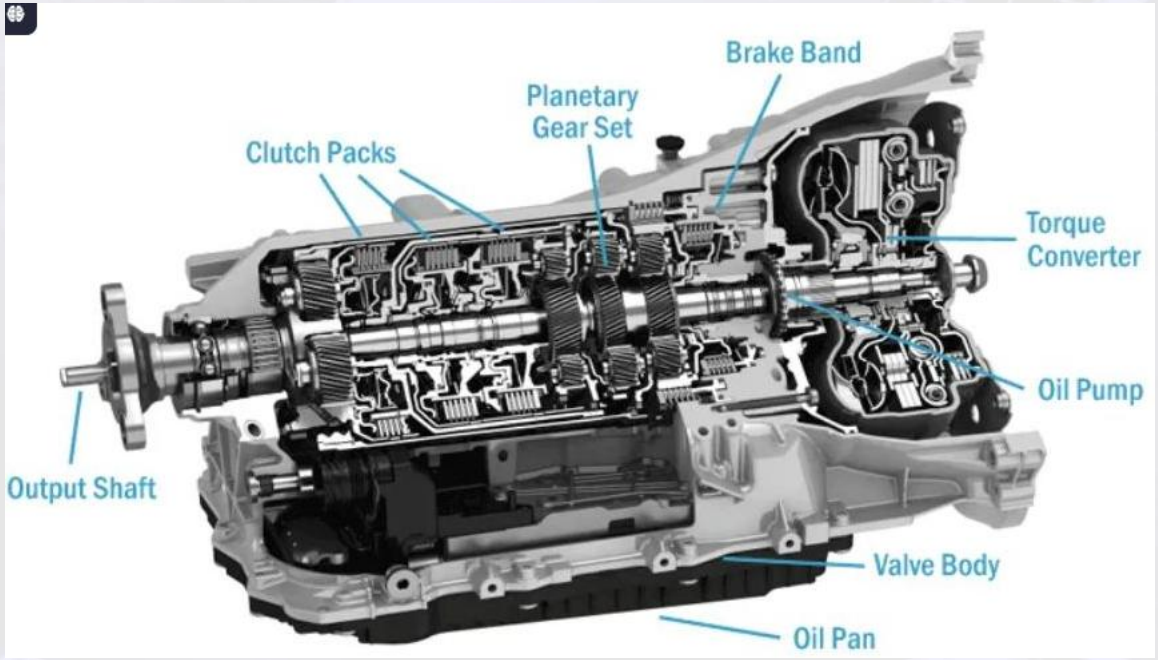
Automatic gear box

An Automatic Transmission (AT) is a vehicle transmission system that automatically changes gear ratios according to vehicle speed, engine load, accelerator position, and driving conditions, without requiring the driver to operate a clutch pedal.



Main Components of an Automatic Transmission

1. Torque converter – transfers engine power to the gearbox.
2. Planetary gear sets – produce different gear ratios.
3. Clutches and brakes – control the motion of planetary gear members.
4. Hydraulic system and valve body – direct pressurized transmission fluid.
5. Transmission Control Unit (TCU) – electronically controls gear shifting.



Torque Converter

The torque converter is a hydrodynamic coupling located between the engine and the automatic gearbox. It allows the engine to continue running while the vehicle is stopped and transfers torque to the transmission through fluid motion.

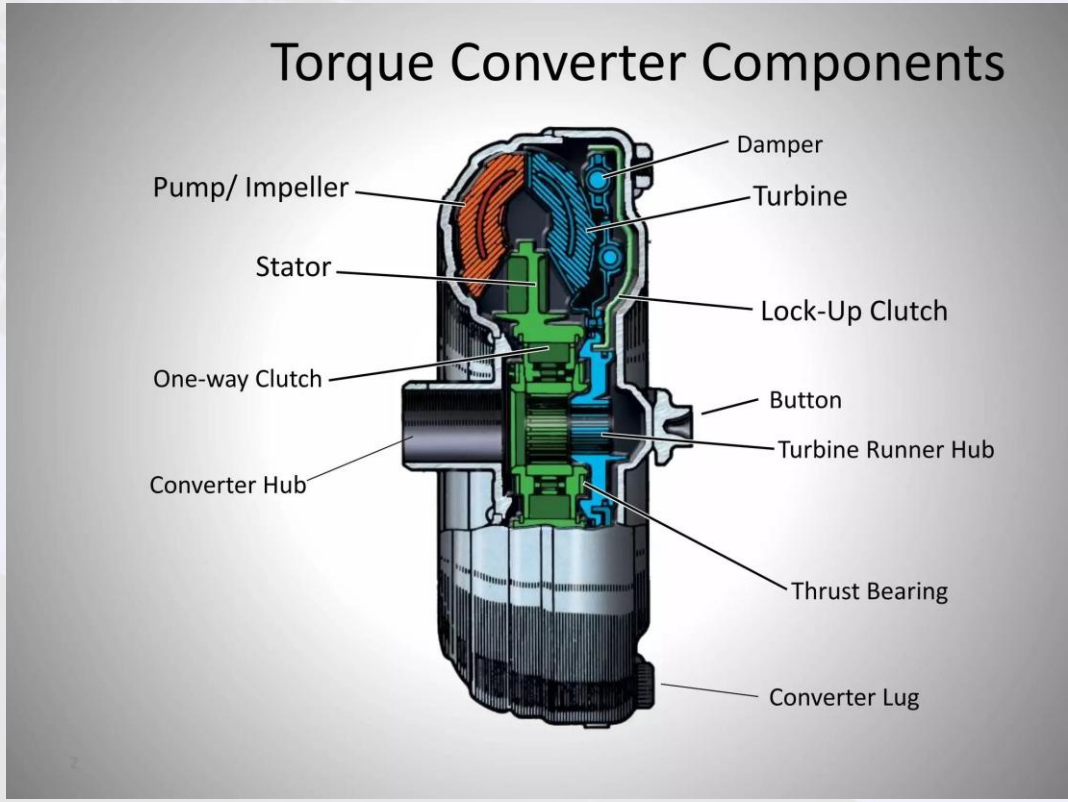
Main components

Impeller (Pump): Connected to the engine and accelerates transmission fluid.

Turbine: Connected to the transmission input shaft and receives torque from the moving fluid.

Stator: Redirects returning fluid to improve torque transfer and provide torque multiplication.

Lock-up clutch: Mechanically connects the engine and transmission input under suitable operating conditions to reduce slip.



Torque Converter

Step-by-step operation:

The engine rotates the impeller.

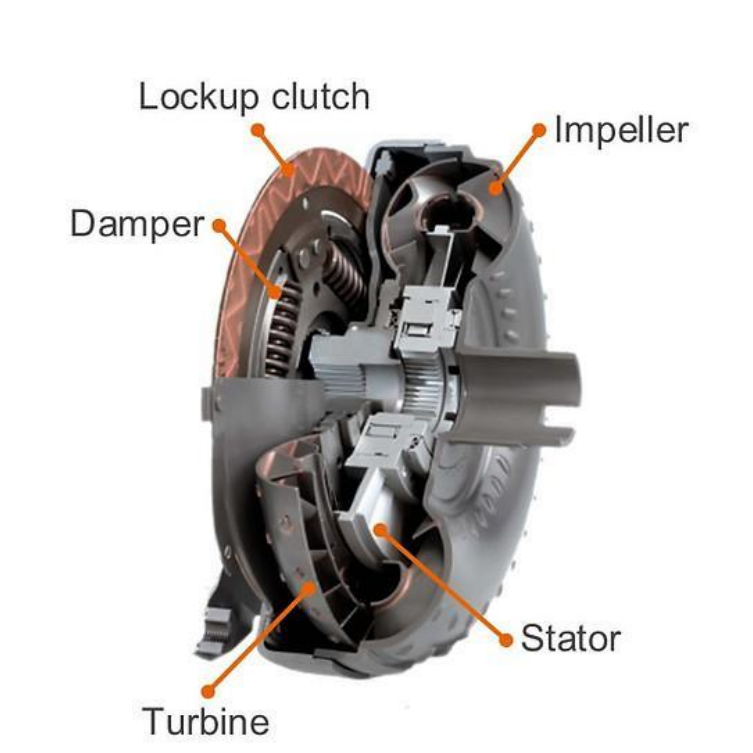
The impeller accelerates the transmission fluid.

The moving fluid strikes the turbine blades, producing torque.

The turbine rotates the transmission input shaft.

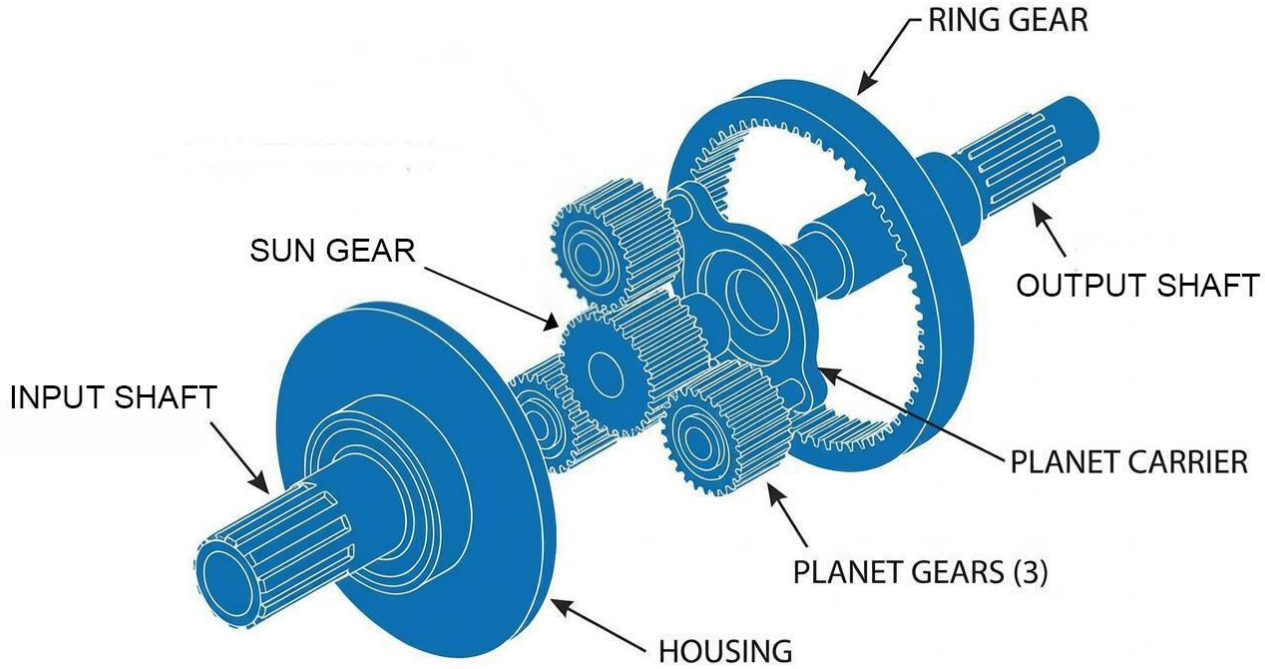
The fluid returns through the stator, which redirects it during torque multiplication.

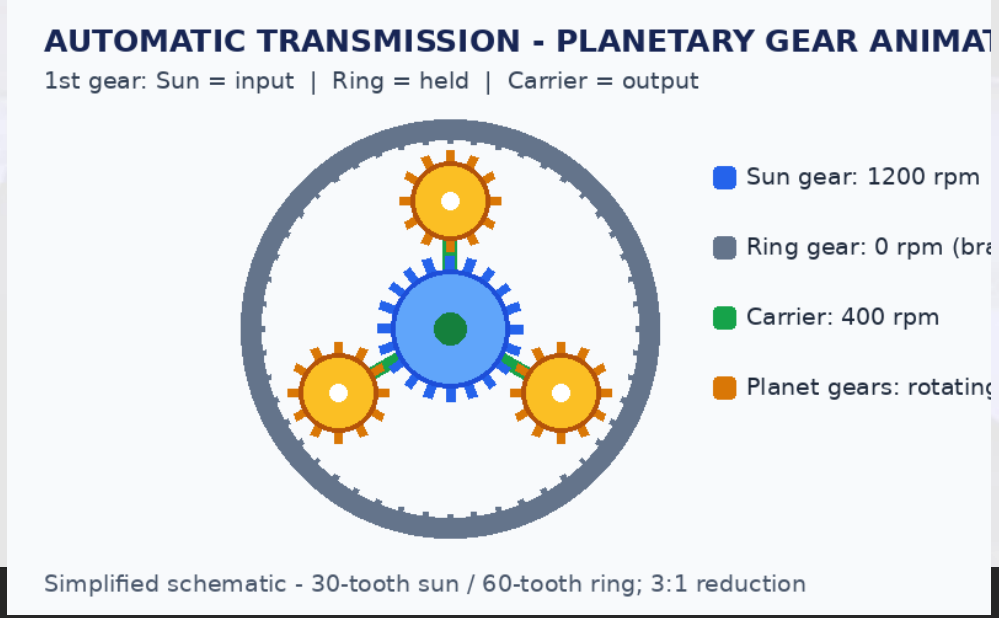
At appropriate cruising conditions, the lock-up clutch can engage, reducing energy losses.



Planetary Gear Set

The planetary gear set is responsible for generating different gear ratios. Instead of sliding gears into and out of mesh, the transmission changes which planetary components are driven, held stationary, or connected to the output shaft.





Automatic gear box

Operating steps:

1. The engine produces rotational power.
2. The clutch connects the engine to the gearbox.
3. When the driver presses the clutch pedal, the engine is disconnected from the transmission.
4. The driver moves the gear lever to select the required gear.
5. The synchronizer matches the rotational speeds of the components before engagement.
6. The driver releases the clutch, and power is transmitted to the wheels.

Advantages: Simple construction, high efficiency, low manufacturing cost, and direct driver control.

Disadvantages: Requires manual operation and can be tiring in heavy traffic.

Continuously Variable Transmission (CVT)

Working Principle

A Continuously Variable Transmission changes the transmission ratio smoothly without using a fixed sequence of gear ratios.

A common automotive CVT uses two adjustable pulleys connected by a steel belt or chain.

Main components:

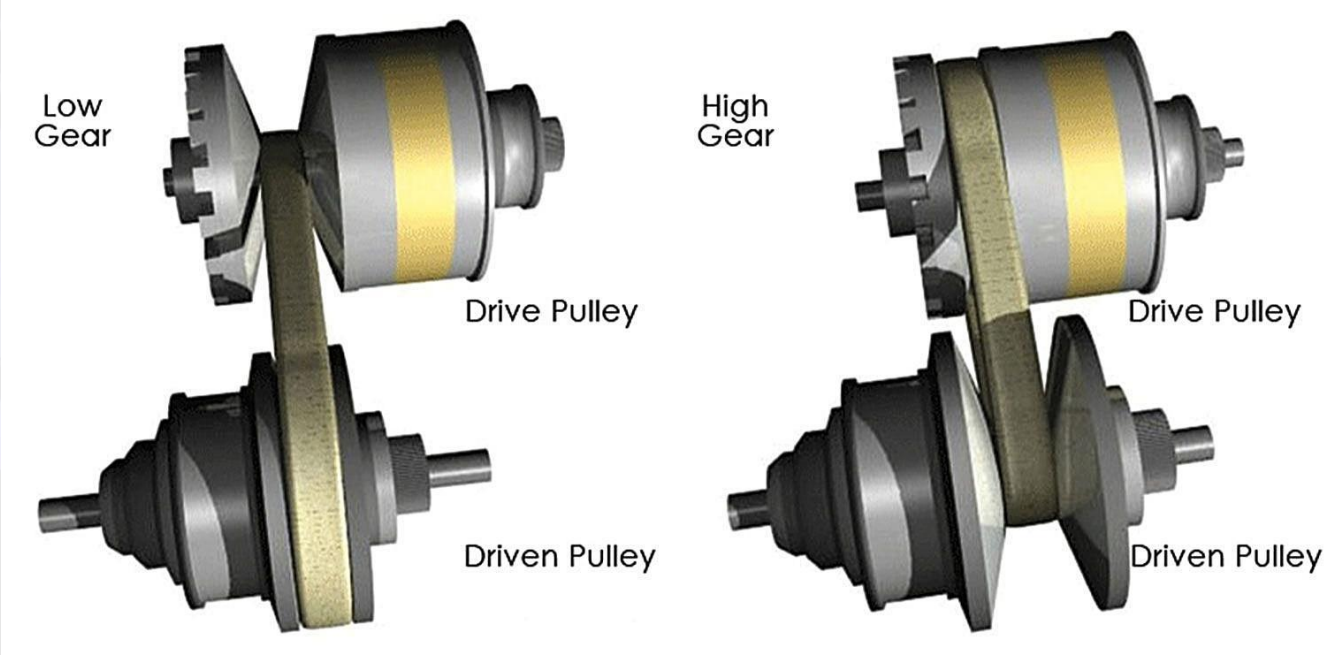
Primary pulley (driving pulley)

Secondary pulley (driven pulley)

Steel belt or chain

Hydraulic or electromechanical pulley control

Transmission Control Unit (TCU)



Continuously Variable Transmission (CVT)

Operating steps

The engine rotates the primary pulley.

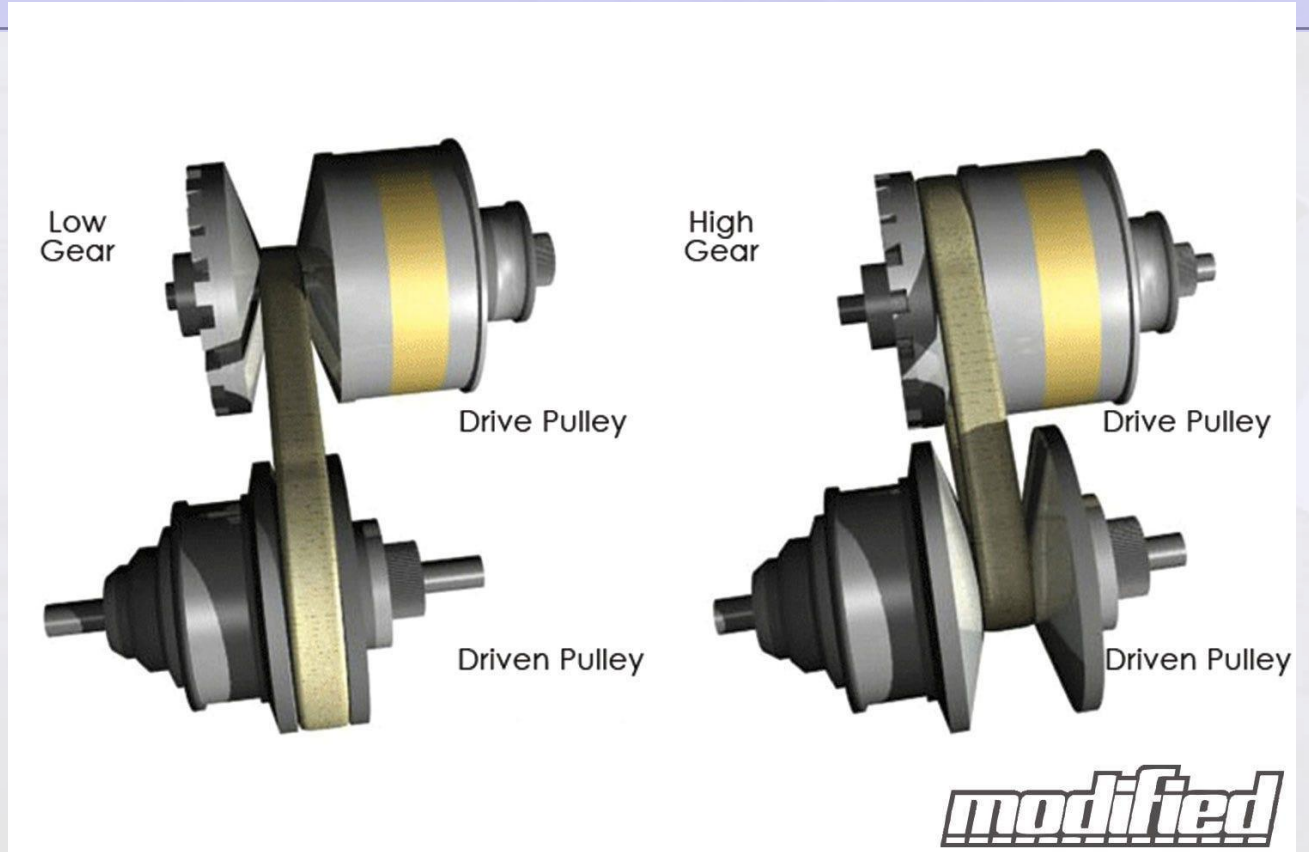
The belt transfers power to the secondary pulley.

The effective diameters of both pulleys are adjusted.

Changing the pulley diameters changes the transmission ratio.

The control system continuously adjusts the ratio according to vehicle speed and engine load.

Continuously Variable Transmission (CVT)



Low-speed condition

Small driving pulley diameter and large driven pulley diameter produce a high speed-reduction ratio and greater output torque.

High-speed condition

Large driving pulley diameter and small driven pulley diameter produce a lower reduction ratio and higher output speed.

Continuously Variable Transmission (CVT)

Advantages: Smooth acceleration, continuously adjustable ratios, and the ability to maintain efficient engine operating speeds.

Disadvantages: Belt or chain loading limitations, specialized maintenance requirements, and a driving feel that differs from fixed-gear transmissions.

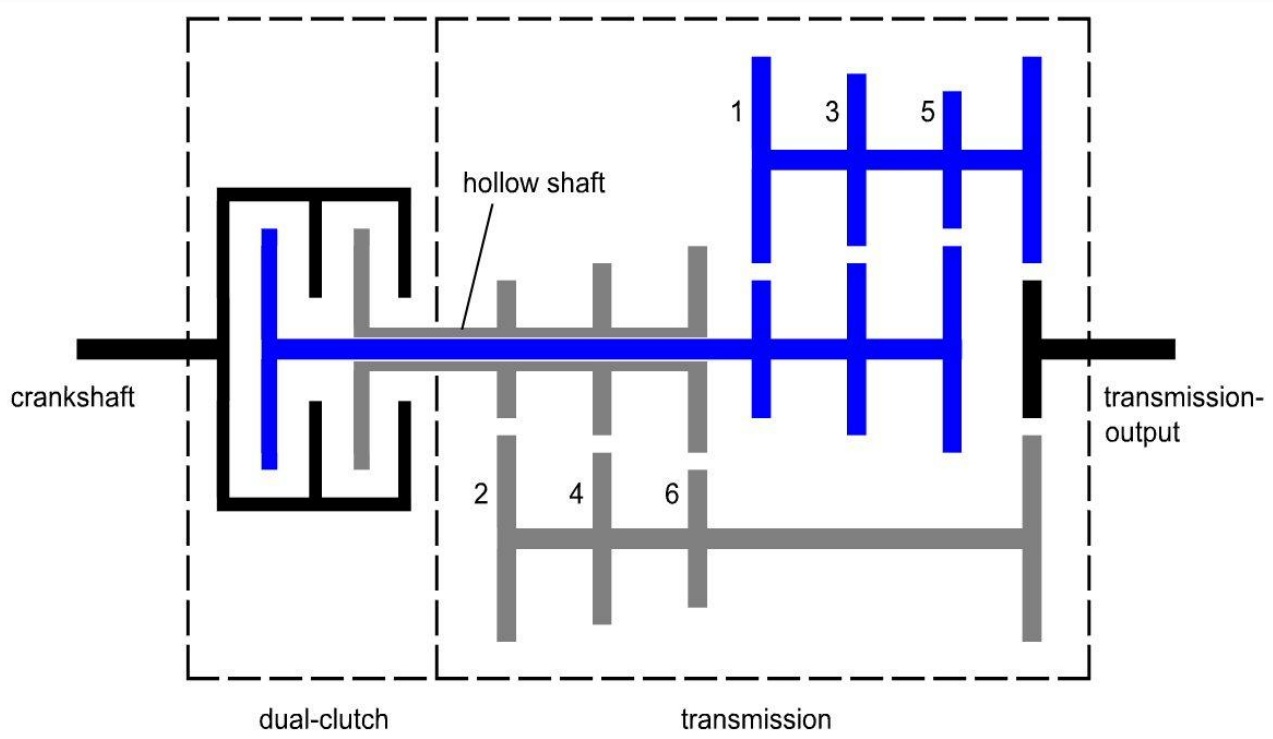
Dual-Clutch Transmission (DCT)

Working Principle

A Dual-Clutch Transmission uses two independently controlled clutches to transmit engine power through two sets of gears.

Typically:

- **Clutch 1:** Controls odd-numbered gears (1, 3, 5, 7).
- **Clutch 2:** Controls even-numbered gears (2, 4, 6).



Dual-Clutch Transmission (DCT)

Operating steps

1. The vehicle moves in first gear through Clutch 1.
2. Second gear is preselected on the other gear shaft.
3. When the control unit requests an upshift, Clutch 1 begins disengaging.
4. Simultaneously, Clutch 2 begins engaging.
5. Engine torque transfers from the first gear path to the second gear path.
6. Third gear can then be preselected for the next shift.

This overlapping clutch operation enables very fast gear changes with minimal interruption of wheel torque.

Advantages: Rapid shifting, high mechanical efficiency, and good acceleration.

Disadvantages: Complex mechanical and electronic control systems, potentially expensive repairs, and possible low-speed hesitation or clutch wear in some designs.

Comparison of Transmission Types

Feature	MT	AT	AMT	CVT	DCT
Gear selection	Driver	Automatic	Automatic	Continuous ratio	Automatic
Clutch pedal	Yes	No	No	No	No
Typical mechanism	Gear pairs	Planetary gear sets	Manual gears with actuators	Variable pulleys	Two gear paths
Shift smoothness	Driver-dependent	Very smooth	Moderate	Very smooth	Generally smooth
Shift speed	Driver-dependent	Fast	Often slower	No discrete shifts	Very fast
Complexity	Low	High	Medium	Medium–high	High
Main advantage	Driver control	Comfort	Low-cost automation	Smooth ratio variation	Rapid gear changes



Thank You!